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NPS/003/003 – Technical Specification for Arc Suppression Coils

1. Purpose

This specification details the technical requirements for Arc Suppression Coils (ASCs) for use on the YEDL & NEDL 11kV distribution networks

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
	Draft	Specification for Arc Suppression Coil Systems

2. Scope

This document applies to ASCs for use in the neutral circuit on the 11kV winding of a system transformer at a YEDL or NEDL substation.

3. Technical Requirements

3.1 Overview

ASC earthing is applied in conjunction with a bypass circuit breaker to form a short time operating system.

The ASC system shall consist of an ASC coil that is suitable for outdoor installation and an associated ASC control unit that will be located in the substation control room.

3.2 Technical Specification

3.2.1 Overall

Arc Suppression Coils (ASC) covered by this Code of Practice shall be provided in accordance with the latest version of IEC 60289.

3.2.2 ASC Voltage Rating

The rated voltage shall be 6350V 50Hz AC between the neutral and earth terminals of the main winding.

A facility shall be provided to measure the neutral voltage.

3.2.3 ASC Current Ratings

YEDL and NEDL will determine the current rating required on a site by site basis.

The coil shall be rated for continuous operation at the current rating stated in the Schedule of Requirements.

3.2.4 ASC Auxiliary Winding

A low current, 110V auxiliary winding shall be provided for measuring and control purposes.

A facility shall be provided to monitor the coil position via an analogue output quantity.

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3.2.5 Surface finish

The ASC coil shall not require maintenance for a period of at least 30 years in a polluted/coastal environment according to EN ISO 12944-2 Category C4.

The surface finish colour shall be the same throughout the ASC and its constituent and auxiliary parts. The surface finish shall be a subtle colour and shade that will visually blend in with other equipment in the substation.

3.2.6 Operating Mechanism

Local control of the ASC shall be provided in a lockable cabinet.

Indication of the operational position of the equipment being controlled shall be provided and shall be clearly visible from the local operating position.

A padlockable Local/Remote control selector switch shall be provided in the local control cabinet. This selector switch shall disable the raise and lower local controls when Remote is selected and shall disable the remote raise and lower controls when Local is selected.

The coil shall be capable of adjustment between the rated current and the lowest obtainable current

3.2.7 Insulating Fluid and Coolant System

Where ASCs for use on the YEDL or NEDL distribution networks contain insulating fluid then this fluid shall comply fully with the current version of CE Electric UK Specification NPS/003/019 – Specification for Electrical Insulating Fluids for use in YEDL & NEDL Plant & Equipment. This requirement also applies to any fluid used for first fill and for the flushing of the unit.

Facilities for fluid filling and sampling shall be provided.

Where the fluid preservation system is a free breathing system the breathing pipe shall be terminated with a 1" (one inch) BSP thread to allow the use of a cartridge system of desiccant filters, such as the Brownell Type W system. The pipework shall be arranged to allow these cartridges to be changed easily from ground level. These requirements apply to all fluid systems that form part of the installation.

3.2.8 ASC Controller

The controller shall be capable of automatically tuning the ASC to match network conditions.

The controller user interface shall have the facilities to display: the coil resonance curve and details of the coil position and status.

Event recording facilities shall be available.

Output indication of an earth fault shall be provided.

The capability for switching the tuning on/off via status inputs shall be provided.

The controller shall be capable of self-monitoring and provide an alarm for this purpose. Monitoring of the ASC operating mechanism shall be provided.

All the output contacts shall be user configurable.

Means of local and remote electronic communication with the controller shall be provided.

The controller shall be suitable for siting in a substation environment.

The unit shall be capable of mounting in a 19" rack in a suitable cabinet.

The unit shall be rated for use with 240V ac, 110V dc and 50V dc supply voltages.

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3.2.9 Main Connections

The termination arrangements shall be as specified in the Schedule of Requirements.

Where open terminal bushings are required these shall be in accordance with IEC 60137.

Where cable connections are required then these shall be separable cable terminations in accordance with BS7888.

3.2.10 Auxiliary Supplies, Small Wiring and Ancillary Equipment (substation applications)

Design and application of electrical ancillary equipment shall be in accordance with ENATS 50-18.

The ASC coil motor supply (if appropriate) shall be 415/240V 50Hz.

All fuses and links shall be in accordance with ENATS 50-18 having labels mounted adjacent to the relevant fuse or link. The label inscription shall include the fuse current rating.

In any control panel or connection chamber; terminals that cannot be isolated electrically shall be fitted with shrouds to prevent the possibility of electric shock by accidental contact.

4. References

4.1 Internal References

Reference	Title
NPS/003/019	Specification for Electrical Insulating Fluids for use in YEDL & NEDL Plant & Equipment

4.2 External References

Reference	Title
IEC 60289	Reactors
ENATS 50-18	Design and Application of Ancillary Electrical Equipment

5. Definitions

Term	Definition

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant & Switchgear Standards Manager	D Blackledge	09/02/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	09/02/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D Gazda	Standards Co-ordinator	D Gazda	09/02/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	24/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	10/02/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	20/02/2006

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Appendix 1 - Schedule of Requirements

Clause	Description	Requirements
3.2.2	Rated Voltage (V)	6350V 50 Hz
3.2.3	Continuous Rated Current (A)	
3.2.9	Termination type (a) Open terminal bushings (b) Cable connection Number & size of cable(s)	